

End-to-End Business Analysis Report
Domain Design · Schema · Features · System Architecture · Auth

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As a Senior Business Analyst, this document presents a comprehensive Travel Agency Domain design — covering end-to-end flow, schema design, features, and enhanced workflow. New, forward-thinking features beyond a standard travel platform have been added to make this a modern, competitive product.

1. System Overview — Architecture Layers

The Travel Agency platform is built in five horizontal layers — from the traveller-facing interface down to data and infrastructure. Each layer has a clear responsibility and communicates only with its adjacent layer.

Layer	Responsibility
Traveller (User Layer)	Browse packages · Book trips · Manage itinerary · Make payments · Leave reviews
Core Application Services	Auth · Search & Filter · Booking Engine · Payment Gateway · Notifications · Itinerary Build
Domain Services	Packages, Bookings, Itinerary, Reviews, Loyalty, Destinations, Vendors, Agents
Third-Party Partner APIs	Airline GDS · Hotel APIs · Payment Processors · Map Services · SMS/Email Gateways
Admin / Operations Layer	Package Mgmt · Booking Control · Vendor Mgmt · Reports · Refund Processing
Data & Infrastructure	PostgreSQL · Redis Cache · S3 Media Storage · Elasticsearch · Message Queue

2. Schema Design — Core Entities

Each entity below represents a MongoDB collection (or PostgreSQL table). Fields are listed in the order they appear in the schema. FK = foreign key reference.

User

```
id, first_name, last_name, email, phone, password_hash,
profile_photo, date_of_birth, nationality, passport_number,
passport_expiry, preferred_currency, preferred_language,
loyalty_points, loyalty_tier (bronze/silver/gold/platinum),
is_verified, is_active, created_at, updated_at
```

Package (the product — a travel package)

```
id, title, slug, description, highlights[], thumbnail,
images[], category_id, destination_id, duration_days,
duration_nights, difficulty_level, max_group_size,
min_age, base_price, discount_price, tax_percent,
is_featured, is_best_seller, is_new,
status (draft/active/inactive), cancellation_policy,
inclusions[], exclusions[], meta_title, meta_description
```

PackageVariant (room type, seat class, meal plan)

```
id, package_id, variant_name,
variant_type (room_type/seat_class/meal_plan), variant_value,
extra_price, available_quantity, images[]
```

Destination

```
id, name, slug, country, state, city, description,
thumbnail, images[], latitude, longitude,
best_time_to_visit, visa_required, visa_info,
weather_info, currency_info, language, time_zone
```

Category

```
id, name, slug, description, thumbnail, parent_category_id,
is_active, sort_order
(e.g. Adventure, Honeymoon, Family, Business, Group Tour, Cruise)
```

Itinerary (day-by-day plan)

id, package_id, day_number, title, description,
accommodation, meals_included[], activities[],
transport_mode, distance_km, highlights[]

Booking (the order)

id, booking_ref, user_id, package_id, variant_id,
travel_date_start, travel_date_end, num_adults,
num_children, num_infants, traveller_details[],
total_amount, discount_amount, tax_amount,
coupon_code, coupon_discount, net_amount,
payment_status (pending/partial/paid/refunded),
booking_status (pending/confirmed/cancelled/completed),
cancellation_reason, refund_amount, refund_status,
special_requests, created_at, updated_at

TravellerDetail (per passenger in a booking)

id, booking_id, first_name, last_name, date_of_birth,
gender, passport_number, passport_expiry,
nationality, is_lead_traveller

Payment

id, booking_id, user_id, amount, currency,
payment_method (card/upi/netbanking/wallet/emi/cod),
gateway (razorpay/stripe/paypal),
gateway_transaction_id, gateway_order_id,
status (initiated/success/failed/refunded),
payment_date, refund_id, refund_date, metadata{}

SavedCard / SavedUPI (stored payment instruments)

id, user_id, card_last4, card_brand, card_expiry,
card_holder_name, gateway_token, is_default

Address (billing address)

id, user_id, address_line1, address_line2,
city, state, country, pincode, is_default

Review

id, booking_id, user_id, package_id,
overall_rating, guide_rating, value_rating,
accommodation_rating, transport_rating,
title, review_text, images[], videos[],
is_verified_purchase, is_published,
admin_reply, created_at

Wishlist

id, user_id, package_id, added_at

Coupon

id, code, description, discount_type (flat/percent),
discount_value, max_discount_cap, min_order_value,
valid_from, valid_to, usage_limit, used_count,
applicable_categories[], applicable_packages[],
is_active

Vendor / Partner (hotel, transport, guide)

id, name, contact_person, email, phone,
vendor_type (hotel/transport/guide/airline),

contract_start, contract_end, commission_percent,
bank_details{}, is_active, rating

Notification

id, user_id, type (booking/payment/offer/system),
title, message, channel (email/sms/push/whatsapp),
is_read, sent_at, metadata{}

LoyaltyTransaction (NEW feature)

id, user_id, booking_id, points_earned,
points_redeemed, transaction_type, balance_after,
expiry_date, created_at

TravelDocument (NEW feature — AI-powered)

id, booking_id, user_id,
document_type (visa/insurance/hotel_voucher/flight_ticket/tour_guide),
file_url, status (pending/generated/delivered), generated_at

3. User Booking Flow — End to End

The following flow covers every step a traveller takes from landing on the website to submitting a post-trip review and claiming loyalty points.

Step	Action	Details
Step 1	Land on Website	Guest or Register / Login (OAuth2 / OTP via SMS or email)
Step 2	Search & Discover	Filter by destination, dates, budget, category, duration, difficulty
Step 3	View Package Detail	Gallery, full itinerary, inclusions/exclusions, reviews, FAQs, map view
Step 4	Select Variant & Dates	Room type, seat class, meal plan, preferred travel dates
Step 5	Enter Traveller Details	Passport number, DOB, nationality per traveller in the group
Step 6	Review & Apply Coupon	Price breakdown, apply loyalty points, enter coupon code
Step 7	Pay Securely	Card / UPI / Wallet / EMI (3-12 months) / Net banking — OTP confirm
Step 8	Booking Confirmed	Booking reference, e-vouchers generated, confirmation via email + SMS
Step 9	Manage Itinerary	Day-by-day plan, travel documents, guide contact, map, hotel details
Step 10	Live Trip Tracking	Real-time guide location on map during active trip
Step 11	Post-Trip Review	Rate guide, hotel, transport, food — upload photos and videos
Step 12	Loyalty Points Credited	Points earned on trip value, tier upgraded if threshold crossed
Step 13	Cancel / Request Refund	Policy-based deduction calculated automatically, refund to source

4. Admin / Operations Flow

4.1 Content Management

- Add Destination — country, visa info, best season, weather, currency info
- Create Package — title, description, price, variants, images, highlights
- Build Day-by-Day Itinerary — activities, meals, accommodation, transport mode
- Set Dynamic Pricing — base price, discount, seasonal rules, demand multiplier
- Manage Coupons — promo codes, flat/percent discount, validity, usage limits
- Publish / Schedule — move from Draft to Active with optional go-live date

4.2 Booking Operations

- View Bookings — filter by status, date range, agent, destination, payment
- Update Status — Pending → Confirmed → In-Progress → Completed
- Assign Resources — tour guide, transport vehicle, hotel room per booking
- Handle Cancellations — policy check, deduction calculation, refund initiation
- Special Requests — dietary needs, accessibility, room preferences
- Reports & Analytics — revenue, occupancy, top destinations, conversion rate

4.3 Booking Status State Machine

Status	Triggered By	Next Possible States
Pending	User submits booking form	Confirmed, Cancelled
Confirmed	Payment successful	In-Progress, Cancelled
In-Progress	Travel date reached	Completed
Completed	Return date passed	— (terminal)
Cancelled	User or admin cancels	Refund Initiated
Refund Init.	Admin approves refund	Refund Done
Refund Done	Gateway processes refund	— (terminal)

4.4 Partner & Finance Operations

- Add Vendor — hotel, transport, guide, airline; link to packages
- Commission Management — auto-ledger, scheduled payouts, commission percent per vendor
- Inventory Control — seats and room availability per date per package
- Send Notifications — email, SMS, WhatsApp, push notifications to travellers
- Review Moderation — approve, publish, reply to, or flag abusive reviews
- CRM — customer history, re-engagement campaigns, loyalty tier management

5. System Design — Tech Stack & Architecture

Layer	Technology
Frontend	Next.js (web) + React Native (mobile)
Backend	Node.js / NestJS (modular monolith or microservices)
Auth	JWT (15 min) + Refresh Token (7 days, HttpOnly cookie)
Primary DB	PostgreSQL
Cache	Redis (sessions, search cache, rate limiting)
Media Storage	AWS S3 / Cloudflare R2
Search	ElasticSearch
Queue	RabbitMQ / BullMQ
Analytics	ClickHouse
Notifications	Firebase (push), Twilio (SMS), SendGrid (email), WhatsApp Business API
Payment	Razorpay / Stripe
Maps	Google Maps API
CDN	Cloudflare

5.1 High-Level Architecture Flow

Client apps (Web, Mobile, Admin Panel, Partner API) hit the API Gateway. The gateway handles JWT validation, rate limiting, CORS, SSL termination, and request logging. Validated requests are routed to the appropriate microservice. Services communicate async events (booking confirmed, payment done, notify trigger) via the Message Queue. All data is persisted in the Data Layer.

Component	Role
API Gateway (Kong / AWS API GW)	JWT validation, rate limiting, CORS, SSL, logging
Auth Service	JWT issuance, OAuth2, OTP, RBAC enforcement
Package Service	CRUD for packages, variants, destinations, categories
Booking Service	Order creation, status transitions, traveller management
Payment Service	Gateway integration, refund processing, saved cards
Notification Service	Email, SMS, push, WhatsApp dispatch via queue
Search Service	ElasticSearch-backed full-text + filtered search
Itinerary Service	Day plan CRUD + AI itinerary generation
Review Service	Rating collection, moderation, admin reply
Loyalty Service	Points earn/redeem, tier calculation, expiry
Message Queue	Async event bus: booking.confirmed, payment.done, notify.trigger

Data Layer	PostgreSQL (core), Redis (cache), S3 (media), ElasticSearch, ClickHouse
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6. Authentication & Authorization

6.1 Auth Flow

- User registers with email/phone → OTP verification (SMS or email)
- Login → server issues JWT access token (15 min) + refresh token (7 days, HttpOnly cookie)
- Social login via OAuth2 — Google, Apple ID
- On checkout → require fresh OTP confirmation for security
- Admin / Agent login → optional 2FA (TOTP app or SMS)
- Vendor partner access → API Key based, rate-limited per key
- Token refresh → silent refresh using refresh token before access token expires
- Logout → refresh token revoked server-side (blacklist in Redis)

6.2 RBAC — Roles & Permissions

Role	Can Do
Guest	Browse packages, view destinations, read reviews
Traveller	All Guest permissions + book, pay, manage own bookings, write reviews, use loyalty points
Travel Agent	All Traveller permissions + book on behalf of clients, view commission dashboard, generate quotes
Vendor	View bookings assigned to them, update availability, upload media, view commission earned
Content Admin	Manage packages, destinations, itineraries, categories, reviews (publish/unpublish/reply)
Finance Admin	View all payments, process refunds, export financial reports, manage coupons
Super Admin	Full access to all modules including user management, role assignment, system config

7. New Features Added (Beyond Standard Travel Platform)

The following 10 features go beyond a typical travel booking platform. They are designed to increase engagement, conversion, and post-trip retention.

1. Loyalty & Rewards Program

Points earned on every booking value. Tier-based perks: Bronze / Silver / Gold / Platinum. Points redeemable against future bookings at checkout. Points expire after 12 months if account is inactive.

2. AI Itinerary Builder

User sets destination + duration + interests (adventure, food, culture, relaxation). AI generates a custom day-by-day plan with hotel suggestions, activities, meal spots. Plan can be saved to account or directly booked as a custom package.

3. Travel Document Vault

All e-vouchers, hotel confirmations, flight tickets, visa docs, and tour guides are stored in one place per booking. Auto-generated and downloadable as a single PDF bundle. Reminders sent 48 hours before travel.

4. Group Booking Management

One lead traveller books for a group. Splits payment link so each member pays their share. Manages all passports and traveller details from a single dashboard. Group discounts applied automatically when group size exceeds threshold.

5. Live Trip Tracking

During active trips, travellers see their tour guide's real-time location on a map. Guide app sends location pings every 30 seconds. SOS button available for emergencies. Family members can be given a read-only tracking link.

6. Flexible Payment — EMI & Pay Later

Split booking cost into 3 / 6 / 12 month EMIs via partner NBFC integrations. Pay Later option: pay 20% now and the rest 30 days before the travel date. Zero cost EMI available for premium loyalty tier members.

7. Dynamic Pricing Engine

Prices fluctuate based on demand, days remaining to departure, and season — similar to airline yield management. Admin sets min/max price bands. Price change history shown to user as a graph to build urgency.

8. Wishlist & Price Alerts

Save any package to wishlist with one click. Set a target price — get notified via email/push when price drops below it. Wishlist shared as a link for couple/family trip planning.

9. B2B Agent Portal

Registered travel agents get their own dashboard with booking history, commission tracking, and white-label quote generation for their clients. Bulk booking discounts and priority support for verified agents.

10. Post-Trip Certificate

Auto-generated 'Trip Completion Certificate' with the traveller's name, destinations visited, photos, and dates. Shareable on social media as a story card. Builds brand awareness organically through user sharing.

8. Complete Enhanced Flow Summary

Admin Side

- Add Destination → Create Package (with variants & itinerary)
- Set pricing (dynamic) → Add coupons → Publish

User Side

- Land → Search (filter by destination/dates/budget/category)
- View package (gallery, itinerary, reviews, FAQs)
- Select variant & dates → Enter traveller details (passport etc.)
- Review checkout (price + loyalty + coupon)
- Pay (card/UPI/EMI/wallet) → OTP confirm
- Booking Confirmed (booking ref + e-vouchers emailed)
- Itinerary dashboard (docs, day plan, guide contact, map)
- Live trip tracking (during trip)
- Post trip: review + loyalty points credited
- Share trip certificate on social media

Admin Post-Booking

- Receive booking → Assign guide + transport + hotel
- Status: Pending → Confirmed → In-Progress → Completed
- Vendor notification via API → user push/SMS notifications
- Handle cancellations (policy-based auto-refund)
- Reports: revenue, occupancy, top destinations, conversion

This is the full end-to-end Travel Agency domain design. Further deep-dives are possible on the AI Itinerary Builder, Dynamic Pricing Logic, ERD Schema Diagram, API Endpoint Design, or any other specific area.